



CYBER SECURITY ESSENTIALS

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AGENDA

- What is Cyber Security?
 - Threats & Risks
 - Cyber Security Organizations
 - Cyber Security Jobs
 - How to get started?
 - Types of Attackers
 - Common Attacks
 - Methods of Attack
 - Defenses against threats
-

COURSE OBJECTIVES

- ❑ We will **NOT** teach you how to be a Hacker.
- ❑ We will teach you a “**Knowledge**”.
- ❑ This knowledge will help you to **protect your systems.**
- ❑ Don't Try to Attack , **without permission.**
- ❑ Try only on sites that **gives Permission.**

قانون حماية البيانات الشخصية - مصر

- يعاقب بالحبس مدة لا تقل عن سنة وبغرامة لا تقل عن مائة ألف جنيه ولا تتجاوز مليون جنيه، أو بإحدى هاتين العقوبتين، كل من :-

- جمع
- أو تداول
- أو عالج
- أو أفصح
- أو أفشى
- أو أتاح

بيانات شخصية بأية وسيلة من الوسائل في غير الأحوال المصرح بها قانونيًا أو بدون موافقة صريحة من الشخص المعني

إجراءات تحرير محضر سب وقذف عبر وسائل التواصل الإجتماعي



- الاتصال بالإدارة العامة للمعلومات والتوثيق - وزارة الداخلية على رقم 0227921490.
- طلب التحويل على ضابط الاتصال وشرح المشكلة.
- التوجه الى إدارة مباحث الإنترنت في المديرية التابع لها
- عمل محضر بالإسكرين شوت للبوست أو الكومنت الذى تسبب فى الإساءة أو الإبلاغ عن الجروب وألاكونت
- التوجه الى مباحث الانترنت فى حالة الفيس بوك ومباحث الاتصالات فى حالة الواتس اب
- يتم تحرير المحضر عن طريق الشاكي فقط أو وكيله الخاص خلال مدة أقصاها 3 أشهر
- تقديم صورة سكرين شوت لما تم من سب أو قذف ويتم إيداع تقرير الفحص الفني يتم إحالة المحضر إلى قسم الشرطة المختص
- يتم ارسال المحضر الى النيابة العامة لاتخاذ الإجراءات القانونية.

What is Security?



“The state of being secure, to be free from danger”

Security is Layers

Layers of Security:

- ❑ **Physical Security:** Protect the **Physical items**, object or areas from unauthorized access. Such as : **Servers – Routers**
- ❑ **Personal Security:** Protection to **personal** who authorized to access organization and its operation.
Such as : **System Admins , Security Engineers**
- ❑ **Network Security:** protection of **networking** components, connection and content such as : **switches – wired - wireless**
- ❑ **Communications Security:** Protection of communication media, networking components and content.
Such as : **Data Transmission – type of encryption protocols**
- ❑ **Information Security:** Protection of information and its Critical elements.

INFORMATION SECURITY?

- Information security (**InfoSec**) involves protecting information and systems (Hardware & Software) from unauthorized access, use, disclosure, disruption, modification, or destruction.

- What is the differences between :**

- Data
- Information
- Knowledge



Cyber Security



WHAT IS CYBER SECURITY?

- security standards which enable organizations to practice safe security techniques to minimize the number of successful cyber security attacks.



CYBER SECURITY

Cyber security refers to the practice of protecting systems, networks, and programs from digital attacks.

- **Cyber**

- Infrastructure (Hardware)
- Operating System
- Application
- Data
- Storage
- Network

- **Security**

- Protect them against attacks or danger



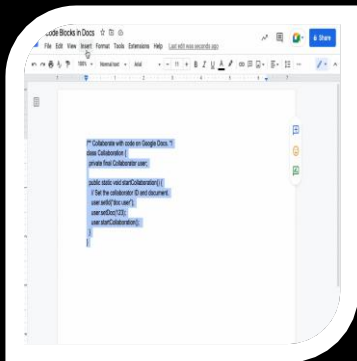
PERSONNEL REPORT

CONFIDENTIAL

Prepared by
MAN Resources Department

Williams, Joan

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WHY DO WE NEED SECURITY?

- **Protect vital information** while still allowing access to those who need it
 - Trade secrets, medical records, etc.
- **Provide authentication and access control for resources**
 - Ex: shared printers/Files/Applications
- **Guarantee availability** of resources
 - Ex: Online services (99.999% reliability)
- **Digitalization** : every thing will be automated

Why Cyber security is important?

- ❑ The Internet is an integral part of our daily lives (24/7).
- ❑ Protects sensitive data (compromised data)
- ❑ Prevents Identity theft
- ❑ Application security
- ❑ Network security
- ❑ **Impact on business and individuals.**
 - ✓ Decrease in productivity and sales revenue
 - ✓ Bad impact on company reputation (yahoo)
 - ✓ Loss of time --→ Money
- ❑ **Increase in Cyber Crimes**

WHAT IS CYBER CRIME?

- "computer crime", or "crime by computer".
- **Cyber Crimes includes**
 - Illegal access
 - Illegal Interception
 - System Interference
 - Data Interference
 - Misuse of devices

Cyber Crime is increasing due to ...

- ❖ **Attack tools evolved-** (no need of sophisticated knowledge).
- ❖ New segment of attackers (Skidi ,WIFI attackers,...)

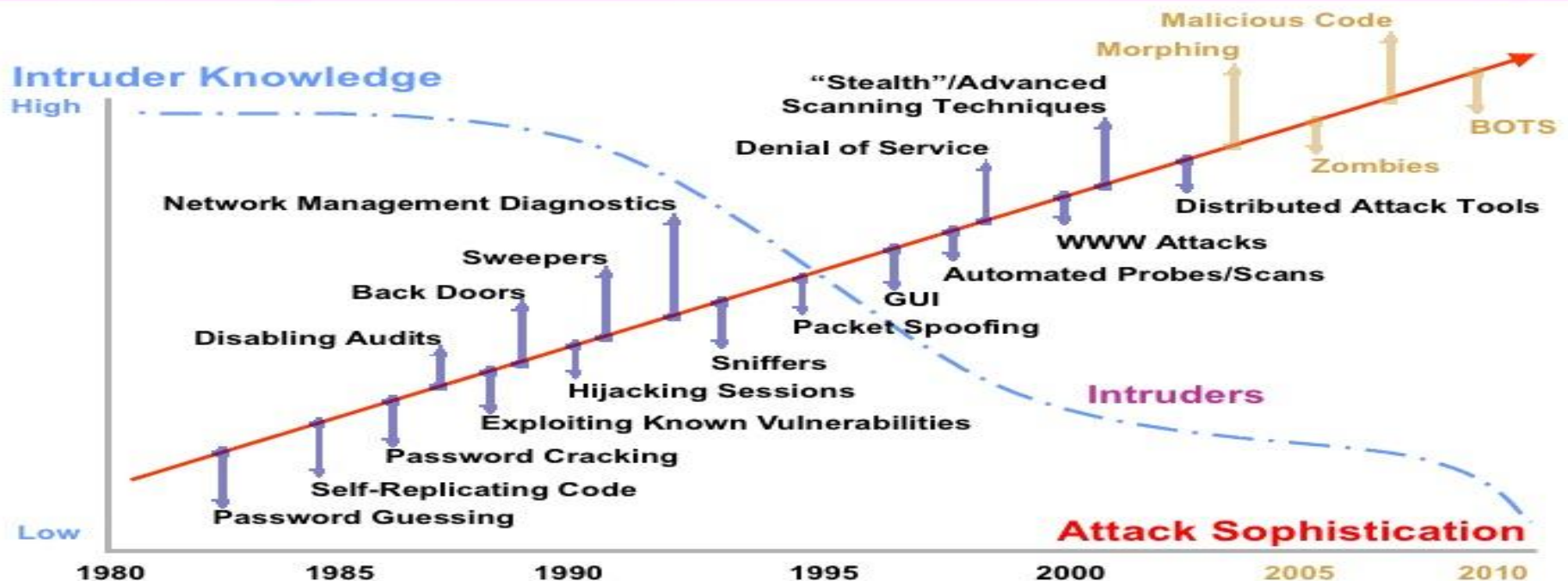
Do you know crimes done by
using Cyber Crimes ??



HISTORY OF CYBER ATTACKS

- The first recorded cyber crime was recorded in the year 1820.
- The first spam email took place in 1978
- The first Virus was installed on an Apple Computer in 1982

Attack Sophistication vs. Intruder Technical Knowledge



FOCUS OF SECURITY IS RISK

- Risk is a **measure** of the cost of realized vulnerability
- Risk is the probability of a **Threat** crossing or touching a **Vulnerability** which casue an **Impact** to the business

$$\text{Risk} = \text{Threat} \times \text{Vulnerabilities} \times \text{Impact}$$

The risk is **High** when

The value of a **vulnerable** asset is **High**

The probability of successful attack (**Threat**) is **High**

The **Impact** to the business is **High**

Vulnerability is the degree of weakness which is inherent in every network and device.

TYPES OF INFORMATION SECURITY THREATS

- **Threats** could be :

A person, thing, event or idea which poses danger to an **asset** in terms of that asset's confidentiality, integrity, availability or legitimate use.

Assets : Everything that have value for an organization or impact its business continuity. This includes personals, hardware, software, physical devices and documents.

Cyber Security is **a shared responsibility**
that each person must accept

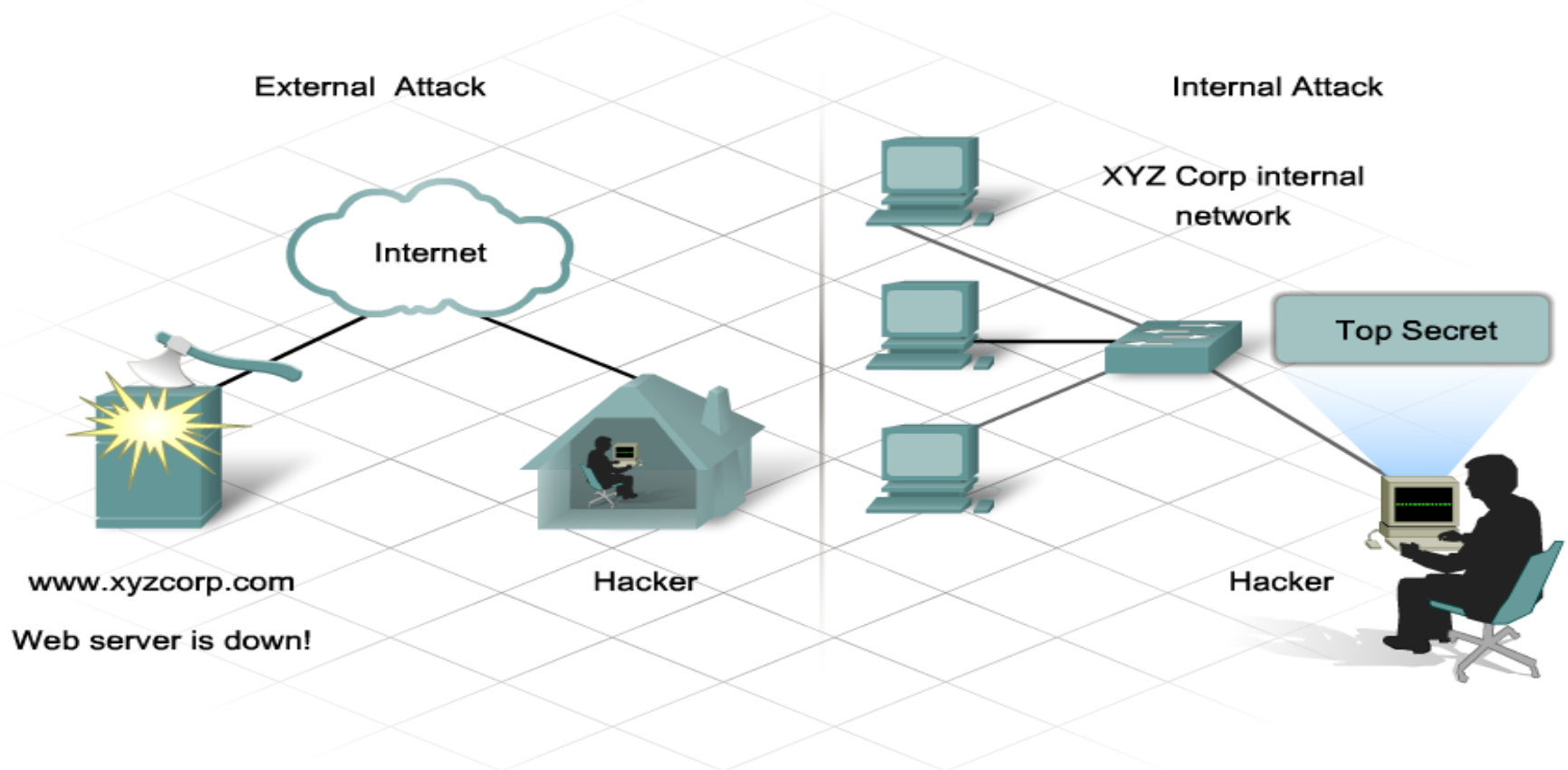
- **Types of Information Security Threats**
 - **Internal Threats**: Threats originating from within the organization, such as **disgruntled employees** or **inadvertent breaches**.
 - **External Threats**: Threats from outside the organization, including **hackers**, **malware**, and **phishing attacks**.
 - **Physical Threats**: Threats to physical hardware and infrastructure, such as **theft**, **natural disasters**,.

EXTERNAL VS INTERNAL THREATS

Which is more dangerous ? Why ?

❑ External

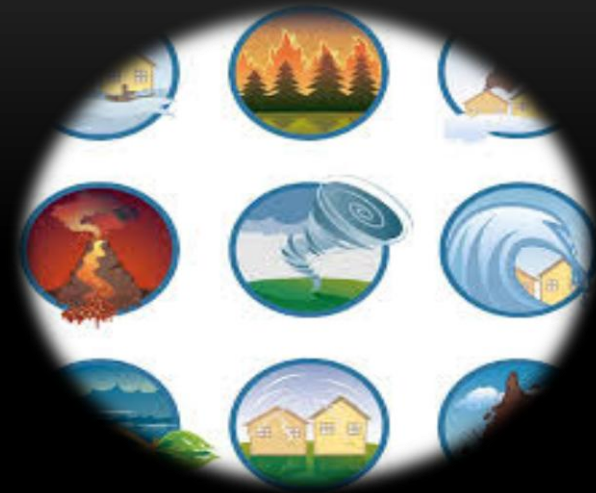
❑ Internal



INFORMATION SECURITY THREATS COULD BE :

- **Accidental** (How to avoid ?)

- Natural disasters
- Revolutions
- Force majeure



- **Intentional** (How to avoid ?)

- Passive (No change)
- Active (Change)



EXAMPLES OF THREATS

- **Information Theft**

- Breaking into a computer to obtain confidential information. Information can be **used** or **sold** for various purposes. **Exams**

- **Data Loss and Manipulation**

- Breaking onto a computer to destroy or alter data records. **(Results)**

- **Identity Theft**

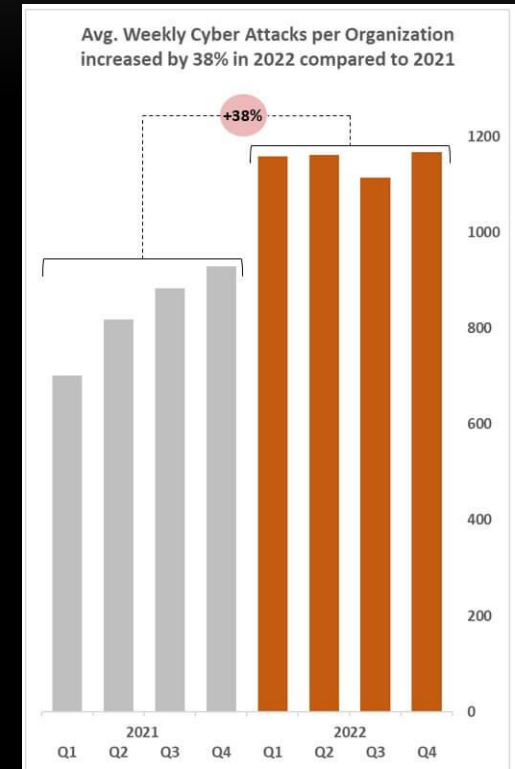
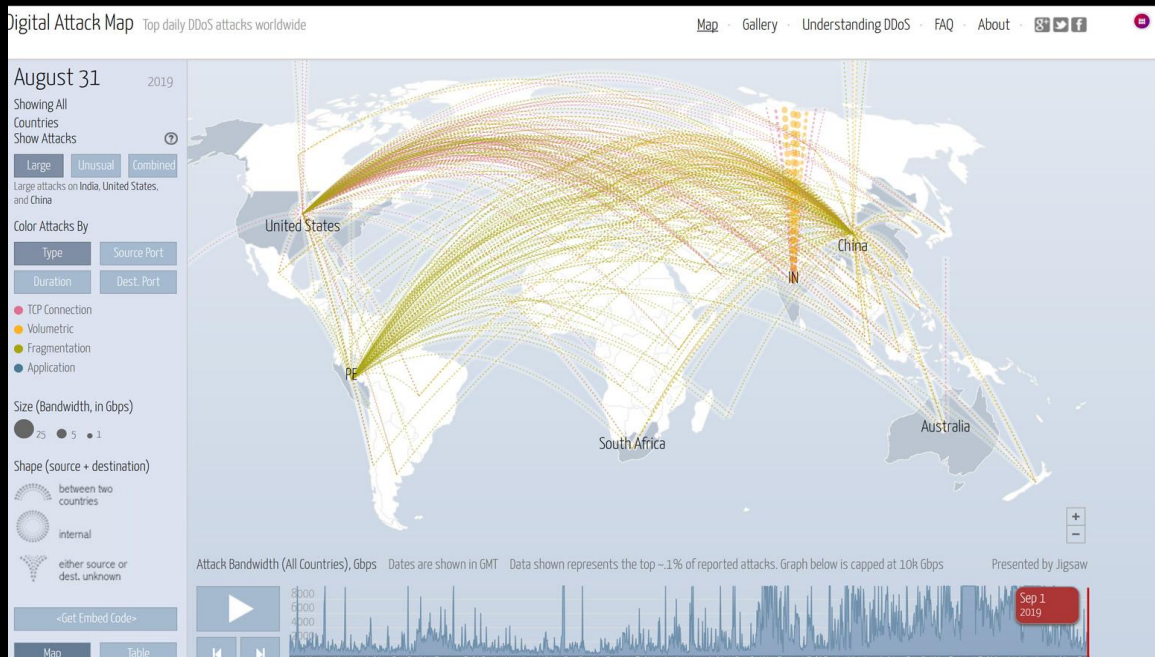
- **Personal Information is stolen** for the purpose of taking over **someone's identity**. Using this information anyone can obtain legal documents, apply for credits and make unauthorized online activities. **Facebook Pages**

- **Disruption of Service**

- Preventing legitimate users from **accessing services** to which they should be entitled **(Home internet access)**.

THREATS ARE INCREASING

- <https://threatmap.fortiguard.com/>
- <https://threatmap.checkpoint.com/>
- https://talosintelligence.com/fullpage_maps/pulse
- <https://threatmap.bitdefender.com/>
- <https://horizon.netscout.com/?mapPosition=0.00~0.00~1.00>



IT'S **IMPOSSIBLE** TO TOTALLY ELIMINATE RISK

- There is **NO simple solution** to secure the information
- Security **99.9 % Not found** Why ?
 - This can be seen through the different types of attacks that users face today.
 - New technologies / applications
 - New Vulnerabilities
 - the difficulties in defending against these attacks



WHO IS NOT ALLOWED TO BE VULNERABLE?

- Financial institutions and banks
- Internet service providers
- Pharmaceutical companies
- Government and defense agencies
- Multinational corporations

ANYONE ON THE NETWORK

SECURITY GOALS IN DIFFERENT ENVIRONMENTS

- Banking
 - Electronic trading
 - Pharmaceuticals
 - All networks
-

SECURITY GOALS IN DIFFERENT ENVIRONMENTS

- **Banking**

- Protect against accidental modification of transactions
- Protect account numbers from disclosure
- Ensure customers privacy

- **Electronic trading**

- Assure source and integrity of transactions
- Protect corporate privacy
- Provide legally binding electronic signatures on transactions

SECURITY GOALS IN DIFFERENT ENVIRONMENTS

- **Pharmaceuticals**
 - Protect corporate / individual privacy
 - Confidentiality is most critical
- **All Networks**
 - Prevent outside penetrations

CYBER SECURITY ORGANIZATIONS



Cyber Security Organizations

➤ www.sans.org



➤ www.cert.org



➤ www.cisecurity.org

➤ www.isc2.org



➤ www.first.org



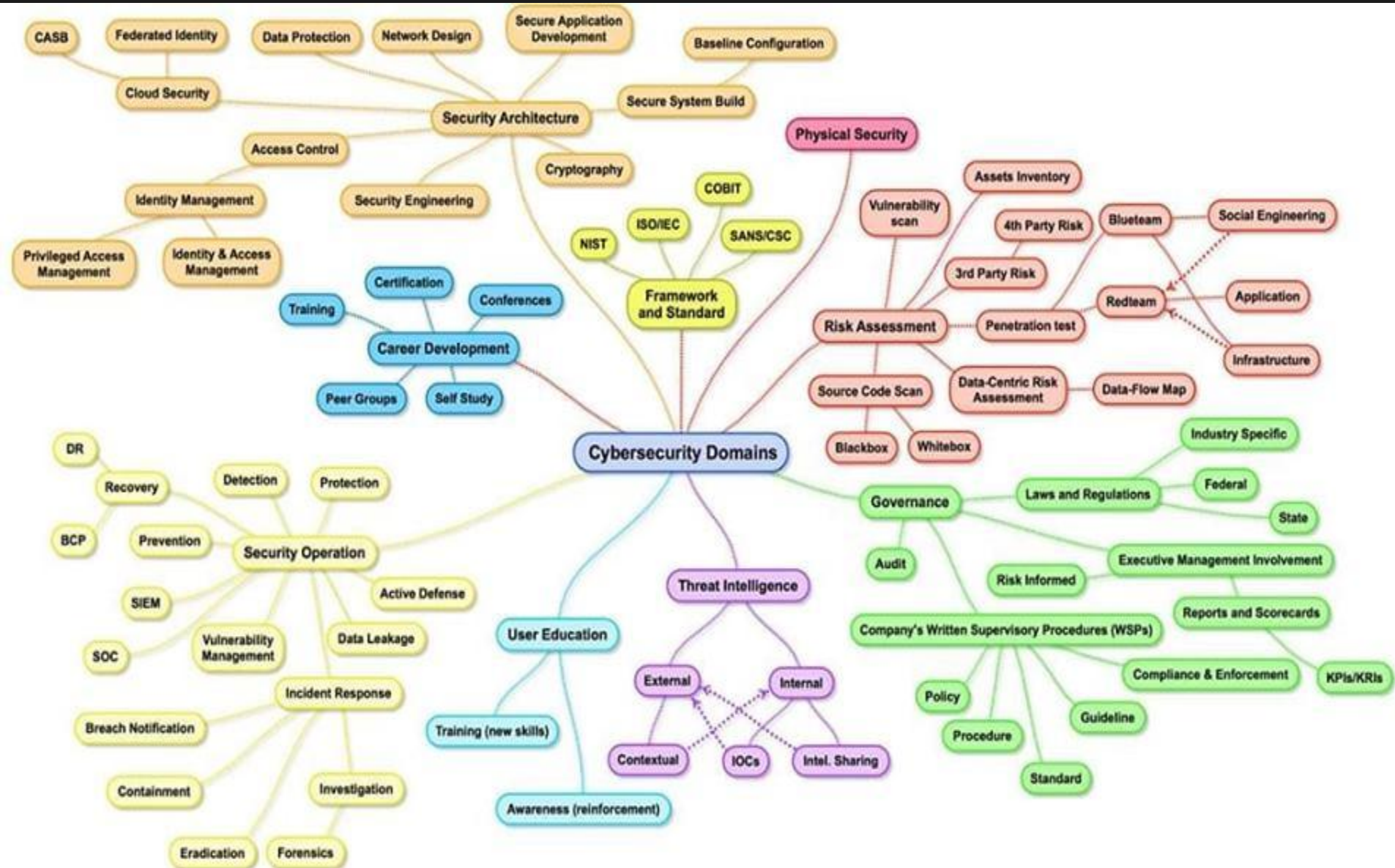
MAJOR CYBER SECURITY SPECIALTIES



MAJOR CYBER SECURITY SPECIALTIES

- System Security
 - Network Security
 - Application Security
 - Penetration Testing
 - Threat / Bug Hunting
 - Malware analysis
 - Digital Forensic
-

CYBER SECURITY DOMAINS (FULL MAP)



CYBER SECURITY JOBS

CYBER SECURITY JOBS TYPES

- **Red Team (Pentest)**

- Vulnerability Assessment.
- Network Pentesting.
- Web Pentesting.
- Mobile Pentesting.
- Wireless Pentesting.
- Advanced Exploitation

- **Blue Team (SOC)**

- Secure Network Design.
- OS Hardening.
- Network Hardening.
- Monitoring.
- Intrusion Detection.
- Incident Response.
- Digital Forensics.
- Malware Analysis.
- Threat Intelligence.
- Threat Hunting.

WHAT IS PENETRATION TESTING ?

- **Pentesting**, known as “**penetration testing**”,
 - security assessment, simulated attacks on an application (web, mobile, ...etc) or network to check its security posture.
- **Team Responsibilities**
 - A highly skilled team
 - looking for vulnerabilities.
 - Looking for unauthorized gaining access to the system's features and data
 - **Simulated** cyberattack but have authorization to do that

RED TEAMING VS PENETRATION TESTING

Aspect	Red Teaming	Penetration Testing
Objective	Test detection and response capabilities	Identify and exploit specific vulnerabilities
Scope	Broad and comprehensive	Narrow and predefined
Duration	Long-term (weeks to months)	Short-term (days to weeks)
Approach	Adversarial, mimicking real-world attacks	Methodical, structured testing
Focus	Overall security posture	Technical weaknesses in specific systems
Outcome	Reports on overall security and defense efficacy	Reports on specific vulnerabilities and fixes
Tools	Various, including custom tools	Standardized testing tools

WHAT IS SECURITY OPERATIONS CENTER (SOC)?

- **Security Operations Center (SOC)**
 - **Centralized unit** that deals with security issues on an organizational and technical level.
- **Purpose:** To monitor, detect, respond to, and mitigate security incidents
- **Team Responsibilities**
 - A highly skilled team
 - Defined definitions, policies, processes and utilizing technologies
 - Manage threats and Reduce security risk and respond to cyber emergencies
 - Monitor, Analyse, Correlate & Escalate Intrusion Events
 - Conduct Incident Management and Forensic Investigation
 - Maintain Security Community Relationships and assist in Crisis Operations
 - Protects mission-critical data and assets.
 - Helps provide continuity and efficient recovery

WORKING FIELDS IN EGYPT

WORKING FIELDS (COMPANIES TYPES) IN EGYPT

- **Vendors**

IBM ,TrendMicro , Cisco

- **Integrators**

- Zinad, Security meter ,Secure Misr, Fixed solutions , ITS ,Salic, Equinox

- **Customer's**

- Banks {Qnb ,Alex ,CIB ..etc}
- Telecom {Orange ,Vodafone ,Etisalat , We}
- Gov {Cert , Central Bank,..}

TOP CYBER SECURITY COMPANIES IN EGYPT

- ❖ Redwerk
 - ❖ Worcbox Technologies
 - ❖ SecureMisr
 - ❖ ISEC
 - ❖ Fixed Solutions
 - ❖ Cyberteq
 - ❖ Softlock
 - ❖ Cyberx
 - ❖ DarkEntery
 - ❖ BBI Consultancy
-

SUCCESS STORY

EGYPTIAN CYBERSECURITY STARTUP SECURE MISR ACQUIRED BY US COMPANY CYSIV

- <https://startups-101.com/2020/03/27/egyptian-cybersecurity-startup-securemistr-acquired-by-us-company-cysiv/>

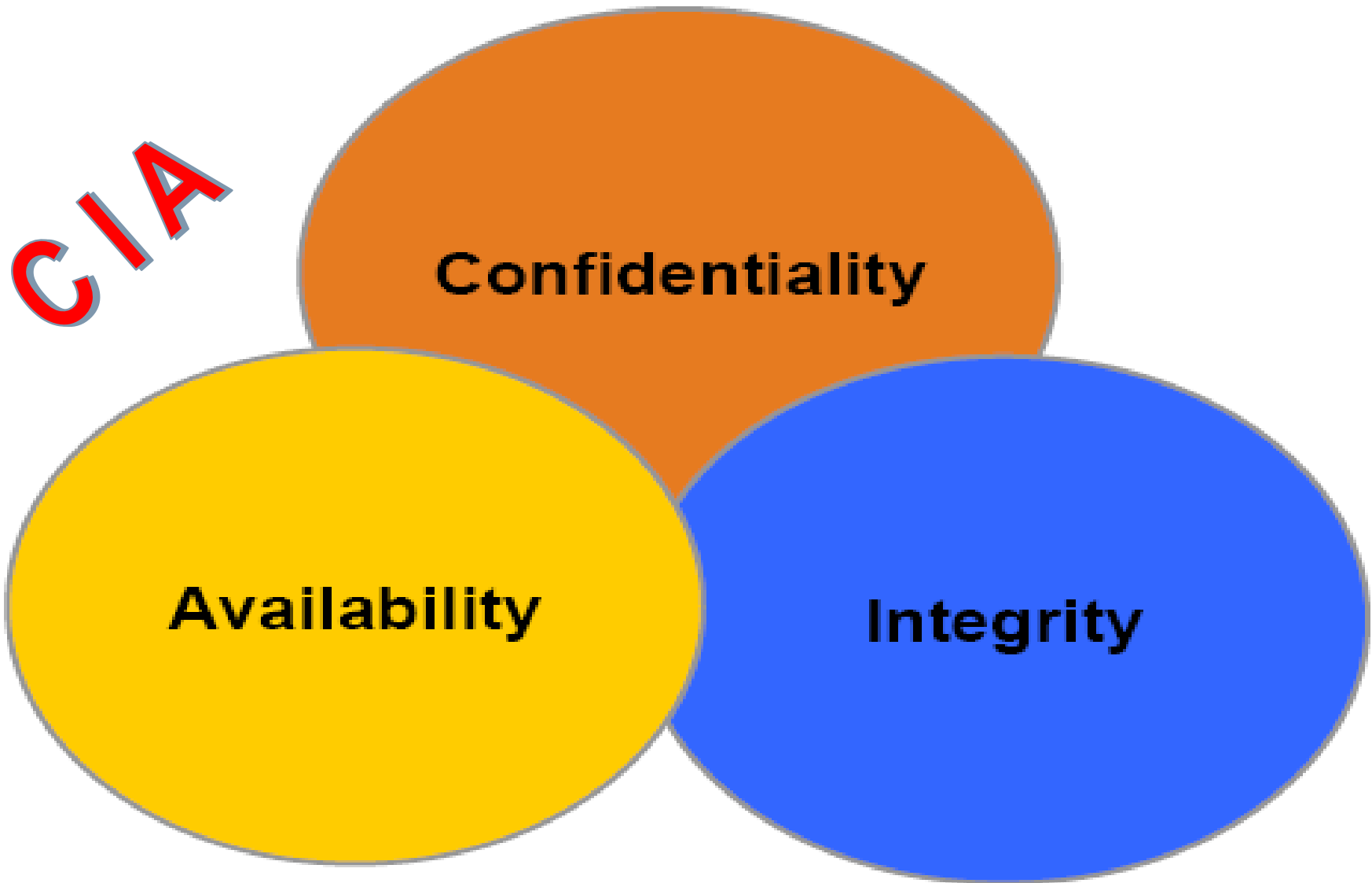


HOW TO GET STARTED



SECURITY GOALS

SECURITY GOALS (OBJECTIVES)



CYBER SECURITY GOALS

- **Confidentiality**

- Ensuring that information is accessible **only** to those authorized to have access.
How ?
- Done by encryption

- **Integrity**

- Ensuring consistency of data and maintaining the accuracy and completeness of data (detect any modification) **How ?**
- Done by Hashing

- **Availability**

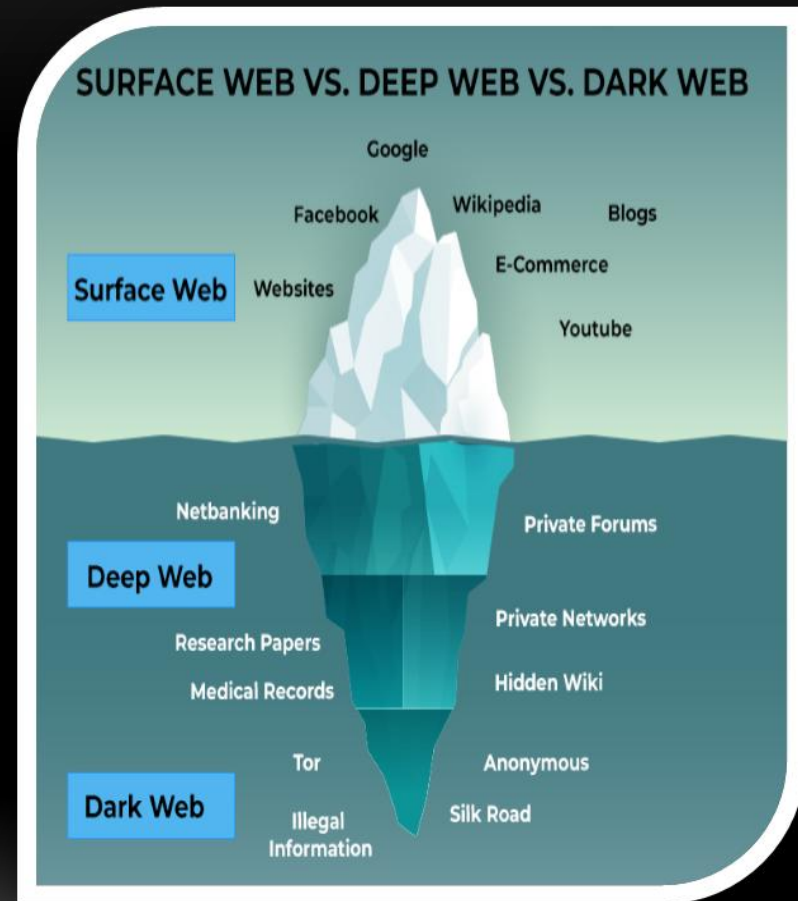
- Ensuring that authorized users have access to information and associated assets when required.**How ?**
- Done by Replications and Redundancy

LAB 1

- Use the hashing tools to validate your hash No.
- https://www.adobe.com/support/adobe_digital_enterprise_platform/trial/md5_checksum.html

SURFACE WEB VS. DEEP WEB VS. DARK WEB

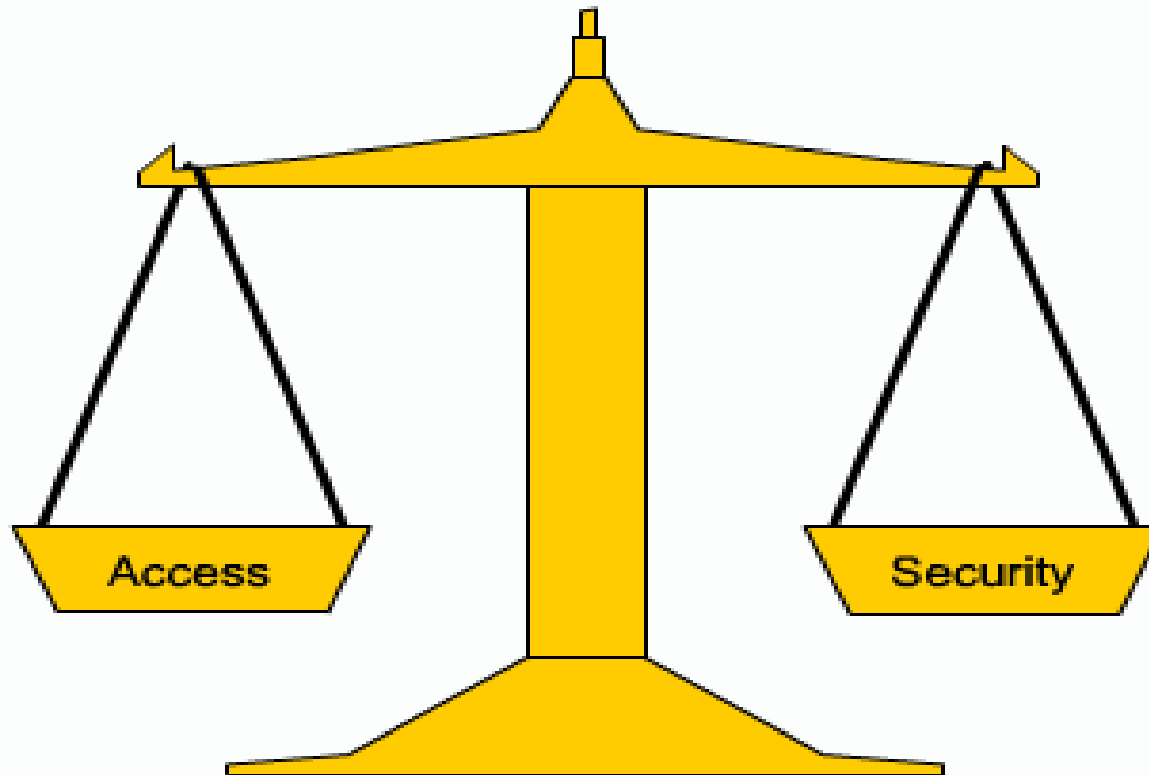
- **Surface Web:** Easily accessible, represents a small fraction of the total web content.
- **Deep Web:** Larger than the surface web, contains valuable and sensitive information, but not inherently malicious.
- **Dark Web:** A small part of the deep web, known for anonymity, often associated with illegal activities but also used for privacy-focused communication.



SURFACE WEB VS. DEEP WEB VS. DARK WEB

Aspect	Surface Web	Deep Web	Dark Web
Definition	Indexed by search engines, publicly accessible	Not indexed by search engines, restricted access	Intentionally hidden, requires special software
Content	Public websites, social media, e-commerce	Private databases, protected sites, subscription content	Anonymous forums, black markets, illicit services
Accessibility	Open to everyone with internet access	Requires passwords, subscriptions, or specific URLs	Requires specific software (e.g., Tor)
Examples	Google, Wikipedia, Amazon, Facebook	Academic databases, online banking, company intranets	Silk Road (defunct), Hidden Wiki, Darknet Markets

CLOSED VERSUS OPEN NETWORKS



Network administrators seek to find a balance between access and security.



Thank You!